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HT Board Booster

Class X Science

Chemistry

CBSE Board Paper Question Bank

Chapter Wise Questions

Ch - 2
Acid, Base and Salts

Session 2026-27



Class X Chemistry
Ch- Acid, Base and Salts
Previous Year CBSE Board Question Paper

CBSE BP 2025-26

Section-A (one Marker Questions)

Q.1. Which one of the following can be used as an acid-base indicator by a visually impaired (blind) student? **(Set- 31/1/1)**

(a) Turmeric (b) Vanilla essence (c) Methyl orange (d) Litmus

Q.2. The natural sources of oxalic acid, lactic acid and methanoic acid respectively are

(a) tomato, curd, ant-sting (b) tomato, orange, nettle-sting **(Set- 31/1/1)**

(c) orange, milk, ant-sting (d) orange, sour milk, nettle-sting

Q.3. The pH of the gastric juices released during digestion is : **(Set- 31/2/1)**

(a) Less than 7 (b) More than 7 (c) Equal to 7 (d) Equal to 0

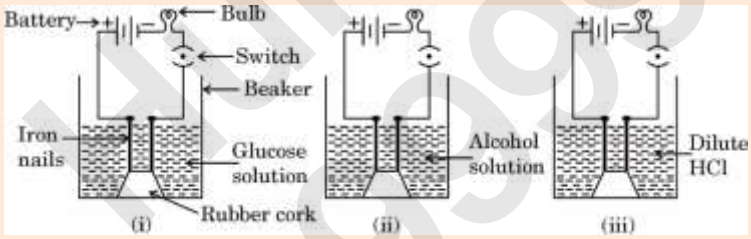
Q.4. Baking powder is a mixture of : **(Set- 31/2/1)**

(a) Sodium carbonate + Acetic acid
(b) Sodium carbonate + Tartaric acid
(c) Sodium hydrogen carbonate + Tartaric acid
(d) Sodium hydrogen carbonate + Acetic acid

Q.5. To a small amount of copper oxide in a beaker, when we add dilute hydrochloric acid slowly with stirring, the change in the colour of the solution is due to the formation of : **(Set- 31/3/1)**

(a) Copper(II) oxide, which is black in colour.
(b) Copper(I) chloride, which is blue in colour.
(c) Copper(II) chloride, which is blue-green in colour.
(d) Copper(I) oxide, which is black in colour.

Q.6.



In which case/cases will the bulb glow considering that all other connections of electric circuit are complete and working ? **(Set- 31/3/1)**

(a) Case (i) only (b) Case (ii) only
(c) Cases (ii) and (iii) (d) Case (iii) only

Q.7. An aqueous solution turns red litmus blue. Excess addition of which of the following solution would reverse the change ? **(Set- 31/4/1)**

(a) Sodium hydroxide (b) Hydrochloric acid
(c) Baking soda (d) Slaked lime

Q.8. How will you protect yourself from the heat generated while diluting a concentrated acid ?

(a) By adding acid to water with constant stirring. **(Set- 31/4/1)**
(b) By adding water to acid with constant stirring.
(c) By adding water to acid followed by base.
(d) By adding base to acid with constant stirring

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Q.9. Study the following table and select the correct option : **(Set- 31/5/1)**

	Salt	Acid used	Base used	Nature of Salt
(A)	NaCl	HCl	NaOH	Basic
(B)	Na_2CO_3	H_2CO_3	NaOH	Neutral
(C)	Na_2SO_4	H_2SO_4	NaOH	Acidic
(D)	CH_3COONa	CH_3COOH	NaOH	Basic

Q.10. Four solutions, namely fructose, ethanol, hydrochloric acid and sodium hydroxide filled in four separate beakers are connected one by one in an electric circuit with a bulb. The solutions in which the bulb will glow when current is passed are : **(Set- 31/5/1)**

(a) ethanol and hydrochloric acid
(b) fructose and ethanol
(c) fructose and sodium hydroxide
(d) hydrochloric acid and sodium hydroxide

Section-C (three Marker Questions)

Q.11. (a) Explain chlor-alkali process with chemical equation. Name the products formed at anode and cathode. **(Set- 31/1/1)**

OR

(b) Write the preparation of following compounds with balanced chemical equation :
(i) Baking soda
(ii) Bleaching powder
(iii) Plaster of Paris

Q.12. Attempt either (a) or (b) : **(Set- 31/3/1)**

(a) Give reasons :
(i) For dilution of an acid, acid is added into water and not water into acid.
(ii) Dry HCl gas does not turn blue litmus red whereas dilute hydrochloric acid does.
(iii) Blue colour of Copper(II) sulphate crystals turns white on heating.

OR

(b) Write the preparation of the following compounds with balanced chemical equation :
(i) NaOH by chlor-alkali process
(ii) Baking Soda
(iii) Plaster of Paris

Q.13. (a) A compound 'A' is used in soda-lime fire-extinguisher and decomposes on heating to form compound 'B'. Compound 'B' is used for removing permanent hardness of water. Identify 'A' and 'B'. **(Set- 31/4/1)**

(b) Write chemical formula of Plaster of Paris. At what temperature gypsum is heated to obtain Plaster of Paris ?

Q.14. Define the following terms : **(Set- 31/5/1)**
(i) Olfactory indicator
(ii) Water of crystallization

Q.15. (A) (a) Give the chemical name and formula of Plaster of Paris. **(Set- 31/5/1)**

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(b) Write the chemical equation of its preparation.

(c) Give any two uses of it.

OR

(B) (a) Name the acid present in ant's sting.

(b) Give reason :

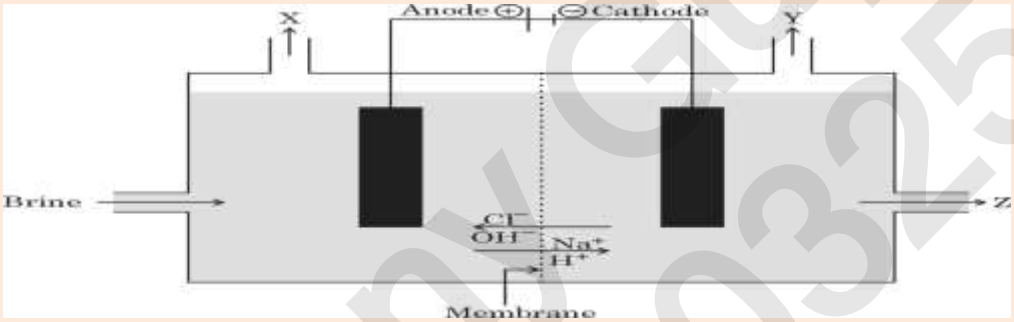
(i) While diluting an acid, it is recommended that the acid should be added to water.

(ii) Baking soda is used as an antacid.

Section-E (five Marker Questions)

Q.16. Read the following passage and answer the questions that follow :

In the given diagram, when electricity is passed through an aqueous solution of a common salt, a substance 'Z' is produced, along with the evolution of gases 'X' and 'Y'. When a burning matchstick is brought near the gas 'Y', it burns with a pop sound, whereas 'X' is used for disinfecting drinking water. When 'X' is passed through slaked lime, an insoluble substance 'A' is produced. **(Set- 31/2/1)**



(a) Write the names of the gases 'X' and 'Y'.

(b) Write the balanced chemical equation for the formation of substance 'Z'.

(c) (i) What happens :

(I) If a drop of red litmus solution is added to the aqueous solution of substance 'Z' ?

(II) If CO_2 gas is passed through ammoniacal solution of aqueous NaCl ?

OR

(c) (ii) Identify the substance 'A' and write the balanced chemical equation of the reaction which takes place in its preparation.

Q.17. The pH plays an important role in our daily life. Strength of acids and bases depends on the number of H ions and OH ions produced by these respectively. pH scale helps in predicting the strength of acids and bases by measuring hydrogen ion concentration in a solution.

Answer the following : **(Set- 31/4/1)**

(a) What type of indicator is used to predict how strong an acid or base is ?

(b) Three solutions A, B and C showed pH as 3, 5 and 7. Which solution has the highest concentration of H ions ?

(c) What are strong acids and weak acids ?

OR

(c) (i) Under what condition is rainwater called acid rain ?

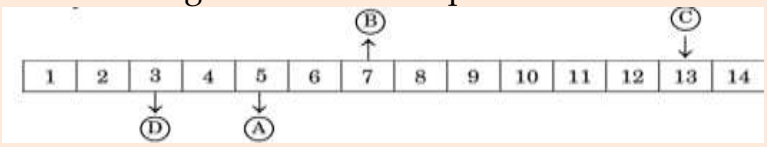
(ii) In the following acids, identify weak acids :

Nitric acid, hydrochloric acid, acetic acid, sulphuric acid, formic acid.

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Section-A (one Marker Questions)

Q.18. The following table shows the pH values of four solutions A, B, C and D on a pH scale: (31/1/1)



The solutions A, B, C and D respectively are of a_____.

(a) Strong acid, weak acid, neutral, strong base
(b) Weak acid, neutral, weak base, strong base
(c) Weak acid, neutral, strong base, weak base
(d) Weak acid, neutral, strong base, strong acid

Q.19. Consider the following reactions: (31/1/1)

(i) Dilute hydrochloric acid reacts with sodium hydroxide.
(ii) Magnesium oxide reacts with dilute hydrochloric acid.
(iii) Carbon dioxide reacts with sodium hydroxide.

It is found that in each case:

(a) Salt and water is formed. (b) Neutral salts are formed.
(c) Hydrogen gas is formed. (d) Acidic salts are formed.

Q.20. The body of human beings works within the pH range of: (31/2/1)

(a) 6.1 to 6.8 (b) 6.5 to 7.3 (c) 7.0 to 7.8 (d) 7.5 to 8.1

Q.21. Which of the given option represents a family of salts? (31/3/1)

(a) NaCl, Na_2SO_4, $CaSO_4$ (b) K_2SO_4, Na_2SO_4, $CaSO_4$
(c) $NaNO_3$, $CaCO_3$, Na_2CO_3 (d) $MgSO_4$, $CuSO_4$, $MgCl_2$

Q.22. In order to prepare dry hydrogen chloride gas in humid atmosphere the gas produced is passed through a guard tube (drying tube) which contains: (31/4/1)

(a) Calcium chloride (b) Calcium oxide
(c) Calcium hydroxide (d) Calcium carbonate

Q.23. **Assertion (A):** Concentrated nitric acid is diluted by adding water slowly to acid with constant stirring. (31/4/1)

Reason (R): Concentrated nitric acid is easily soluble in water.

Q.24. A common feature observed in the crystals of washing soda, copper sulphate, gypsum and ferrous sulphate is that all (31/5/1)

(a) exhibit basic nature
(b) exhibit acidic nature
(c) have fixed number of molecules of water of crystallisation in one formula unit of these salts.
(d) are coloured

Q.25. A metal, 'X', on treatment with sodium hydroxide liberates a gas 'G'. It also liberates the same gas, 'G' on treatment with dilute sulphuric acid. (31/5/1)

Based on above information, 'X' and 'G' respectively are

(a) Copper and Sulphur dioxide (b) Zinc and Sulphur dioxide
(c) Zinc and Hydrogen (d) Copper and Hydrogen

Q.26. You have three aqueous solutions A, B and C as given below: (31/6/1)

A - Potassium nitrate
B - Ammonium chloride
C - Sodium carbonate

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The ascending order of the pH of these solutions is:

(A) A < B < C (B) B < C < A (C) C < A < B (D) B < A < C

Q.27. Juice of tamarind turns blue litmus to red. It is because of the presence of a chemical compound called (31/6/1)

(a) Acetic acid (b) Methanoic acid

(c) Oxalic acid (d) Tartaric acid

Q.28. The water of crystallization is present in (31/6/1)

(i) Bleaching Powder (ii) Plaster of Paris

(iii) Washing Soda (iv) Baking Soda

(A) (ii) and (iv) (B) (ii) and (iii)

(C) (i) and (iii) (D) (i) and (iv)

Section-B (two Marker Questions)

Q.29. (a) Write the formula of the ions which (i) acids, and (ii) bases generate in water solutions. (31/3/1)

(b) Dry HCl gas does not change the colour of dry litmus paper. Why?

Section-C (three Marker Questions)

Q.30. (a) Common salt is an important raw material for various chemicals of daily use. State in brief the method of preparation of (i) Sodium hydroxide, and (ii) Sodium hydrogen carbonate from common salt. Write balanced chemical equations of the reactions that occur. (31/3/1)

OR

(b) Design an experimental set-up to demonstrate that "Alcohol and glucose contain hydrogen but are not categorised as acids". Also give the reason to justify this fact.

Q.31. Write chemical formula of washing soda. How is it obtained from baking soda? List two uses of washing soda. (31/5/1)

Q.32. (a) State the chemical property in each case on which the following uses of baking soda are based upon : (31/6/1)

(i) as an anti-acids

(ii) as a constituent in making baking powder

(iii) in soda-acid fire-extinguishers

OR

(b) Write chemical equations to show what happens when an acid reacts with a

(i) metal (ii) base and (iii) carbonate

Write the name of the main product formed in each case.

Section-E (five Marker Questions)

Q.33. Seawater contains many salts dissolved in it. Common salt is separated from these salts. Deposits of solid salt are also found in several parts of the world. These large crystals are often brown due to impurities. This is called rock salt and is mined like coal. The common salt is an important raw material for chemicals of daily use. (31/1/1)

(a) Write balanced chemical equations to show the products formed during electrolysis of brine.

(b) List two uses of any one product obtained during electrolysis of brine.

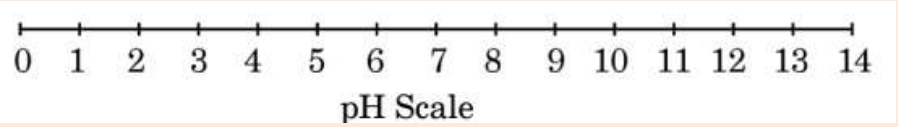
(c) (i) A mild non-corrosive basic salt 'A', used for faster cooking, is strongly heated to produce a compound 'B', that is used for removing permanent hardness of water. Identify A and B and also write the equation for the reaction that occurs when A is heated.

OR

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(c) (ii) Define water of crystallisation. Give two examples of salts that have water of crystallisation.

Q.34. Acid-base indicators can be used to distinguish between an acid and a base. Universal indicator, a mixture of several indicators, shows different colours at different concentrations of acids and bases, thereby indicating their pH on the pH scale of 0- 14. The pH of a solution is measured by pH paper, which is a paper impregnated with a universal indicator.



Answer the following questions: (31/2/1)

Q.35. Solution P is a strong acid while solution Q is a strong base. On the pH scale, where would you place the solutions P and Q?

(b) A solution has a pH of 7. Name a compound you would use to

(i) increase its pH, and (ii) decrease its pH.

(c) (i) When the pH of a solution is decreased from 4 to 2, what effect does it produce on its hydronium ion concentration? State the colour change shown by the pH paper.

OR

(c) (ii) A person is feeling pain and irritation in the stomach due to indigestion. What could be the pH of the fluid in the stomach? Write the common name of the medicines people use for remedy. Give the chemical name of "milk of magnesia" often used for this purpose.

Q.36. Common salt is a very important chemical compound for our daily life. It's chemical name is sodium chloride and it is used as a raw material in the manufacture of caustic soda, washing soda, baking soda etc. It is also used in the preservation of pickles, butter, meat etc.

(i) Name the acid and the base from which common salt can be obtained. (31/4/1)

(ii) State the nature (acidic/basic/neutral) of sodium chloride. Give reason for the justification for your answer.

(iii) (A) What happens when electric current is passed through an aqueous solution of sodium chloride (called brine)? Name the products obtained along with the corresponding places in the electrolytic cell where each of these products is obtained.

OR

(iii) (B) How is washing soda obtained from sodium chloride? Give chemical equation of the reactions involved in the process.

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Section-A (one Marker Questions)

Q.37. Consider the following compounds: (Set- 31/1/1)

$FeSO_4$; $CuSO_4$; $CaSO_4$; Na_2CO_3

The compound having maximum number of water of crystallization in its crystalline form in one molecule is:

(a) $FeSO_4$ (b) $CuSO_4$ (c) $CaSO_4$ (d) Na_2CO_3

Q.38. Solid Calcium oxide reacts vigorously with water to form Calcium hydroxide accompanied by the liberation of heat. From the information given above it may be concluded that this reaction (Set- 31/2/1)

(a) is endothermic and pH of the solution formed is more than 7.
(b) is exothermic and pH of the solution formed is 7.
(c) is endothermic and pH of the solution formed is 7.
(d) is exothermic and pH of the solution formed is more than 7.

Q.39. Juice of tamarind turns blue litmus to red. It is because of the presence of an acid called : (Set- 31/2/1)

(a) methanoic acid (b) acetic acid (c) tartaric acid (d) oxalic acid

Q.40. Identify the product 'X' obtained in the following chemical reaction: (Set- 31/3/1)

$CaCO_3 \xrightarrow{\Delta}$ 'X' + CO_2

(a) Quick lime (b) Gypsum (c) Lime Stone (d) Plaster of Paris

Q.41. Select a pair of natural indicator from the following: (Set- 31/3/1)

(a) Litmus and methyl orange (b) Turmeric and Litmus
(c) Phenolphthalein and methyl orange (d) Methyl orange and Turmeric

Q.42. A chemical compound used in glass, soap and paper industries is (Set- 31/3/1)

(a) Washing Soda (b) Baking Soda
(c) Bleaching Powder (d) Common Salt

Q.43. An aqueous solution of a salt turns blue litmus to red. The salt could be the one Obtained by the reaction of (Set- 31/4/1)

(a) HNO_3 and NaOH (b) H_2S0_4 and KOH
(c) CH_3COOH and NaOH (d) HCL and NH_4OH

Q.44. Four solutions, namely glucose, alcohol, hydrochloric acid and sulphuric acid filled in four separate beakers are connected one by one in an electric circuit with a bulb. The solutions in which the bulb will glow when current is passed are : (Set- 31/4/1)

(a) Glucose and alcohol (b) Alcohol and hydrochloric acid
(c) Glucose and sulphuric acid (d) Hydrochloric acid and sulphuric acid

Q.45. The salt present in tooth enamel is : (Set- 31/5/1)

(a) Calcium phosphate (b) Magnesium phosphate
(c) Sodium phosphate (d) Aluminum phosphate

Q.46. In aqueous solution of sodium chloride is prepared in distilled water. The pH of this solution is : (Set- 31/5/1)

(a) 6 (b) 8 (c) 7 (d) 3

Section-B (two Marker Questions)

Q.47. (a) 1 gram of solid sodium chloride was taken in a clean and dry test tube and concentrated sulphuric acid was added to it. (Set- 31/4/1)

(i) Name the gas evolved in the reaction.

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(ii) What will be observed when this gas is tested with (I) dry, and (II) wet blue litmus paper ? Write your conclusion about the nature (acidic/basic) of this gas.

OR

(b) Some metals react with acids to produce salt and hydrogen gas. Illustrate it with an example. How will you test the presence of this gas ?

Section-C (three Marker Questions)

Q.48. (i) The pH of a sample of tomato juice is 4.6. How is this juice likely to be in taste? Give reason to justify your answer. (Set- 31/1/1)

(ii) How do we differentiate between a strong acid and a weak base in terms of ion-formation in aqueous solutions?

(iii) The acid rain can make the survival how?

Q.49. Write the common name and the chemical name of the compound $CaSO_4 . \frac{1}{2} H_2O$. Write the method of its preparation. Give chemical equation for the reaction, when water reacts with $CaSO_4 . \frac{1}{2} H_2O$. (Set- 31/4/1)

Section-D (five Marker Questions)

Q.50. **(a)** (i) Five solutions A, B, C, D and E when tested with pH paper showed pH as 4, 1, 13, 7 and 10 respectively. Which solution is: (Set- 31/3/1)
(1) Strongly acidic (2) Strongly alkaline (3) Weakly acidic
(4) Neutral and (5) Weakly alkaline? Arrange the solutions in increasing order of H+ ion concentration.

(ii) Write the name and formula of (1) an acidic salt and (2) a basic salt giving the name of the parent acid and parent base used to form the salt in each case.

OR

(b) Name and state in brief the process which is used to prepare sodium hydroxide from sodium chloride. In this process along with the main product two gases 'X' and 'Y' are also given off at the two electrodes. Name 'X' and 'Y' specifying the name of their respective electrode at which each gas is obtained. One of these gases when reacts with dry calcium hydroxide produces a compound 'Z' which is widely used in water treatment plants and textile industries. Name Z and write chemical equation for the reaction involved in its formation.

Q.51. (a) A few crystals of ferrous sulphate were taken in a dry boiling tube and heated. Tiny water droplets were observed in the tube after some time. (Set- 31/5/1)

(i) From where did these water droplets appear? Explain.

(ii) What colour change will be observed during heating?

(iii) How many molecules of water are attached per molecule of $FeSO_4$ crystal ? Write the molecular formula of crystalline forms of (I) Copper sulphate, and (II) Sodium carbonate.

(iv) State how is Plaster of Paris obtained from gypsum. Write two uses of Plaster of Paris.

OR

(b) An Acid 'X' present in tamarind when mixed with 'Y', produce a mixture 'Z'.' Z' on addition to a dough when heated makes cakes soft and spongy. 'Y' is prepared from common salt and helps in faster cooking.

(ii) How is 'Y' prepared and how does it help in making soft and spong? Illustrate the reaction with suitable chemical equation.

(iii) Write the name and chemical formula of a mild base other than 'Y' used as an antacid?

Section-E (four Marker Questions)

Q.52. Salts play a very important role in our daily life. Sodium chloride which is known as common salt is used almost in every kitchen. Baking soda is also a salt used in faster cooking as well as in baking industry. The family of salts is classified on the basis of cations and anions present in them, (Set- 31/2/1)

(a) Identify the acid and base from which Sodium chloride is formed.

(b) Find the cation and the anion present in Calcium sulphate.

(c) "Sodium chloride and washing soda both belong to the same family of salts." Justify this statement.

OR

(c) Define the term pH scale. Name the salt obtained by the reaction of Potassium hydroxide and Sulphuric acid and give the pH value of its

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Section-A (one Marker Questions)

Q.53. The table below has information regarding pH and the nature (acidic/basic) of four different solutions. Which one of the options in the table is correct? (Set- 31/1/1)

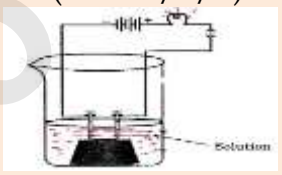
Option	Solution	Colour of pH paper	Approximate pH value	Nature of solution
(a)	Lemon Juice	Orange	3	Basic
(b)	Milk of magnisia	Blue	10	Basic
(c)	Gastric Juice	Red	6	Acidic
(d)	Pure Water	Yellow	7	Neutral

Q.54. Select washing soda from the following: (Set- 31/1/1)

(a) $NaHCO_3$ (b) $Na_2CO_35H_2O$

(c) $Na_2CO_3.10H_2O$ (d) NaOH

Q.55. In the given experimental set-up, if the experiment is carried out separately with each of the following solutions the cases in which the bulb will glow is/are: (Set- 31/2/1)



(i) Dilute hydrochloric acid (ii) Dilute sulphuric acid

(iii) Glucose solution (iv) Alcohol

(a) (i) only (b) (ii) only

(c) (i) and (ii) (d) (ii), (iii) and (iv)

Q.56. In order to balance the following chemical equation, the values of the coefficients x and y respectively are: (Set- 31/2/1)

$x\ Pb(NO_3)_2 \xrightarrow{Heat} 2\ PbO + y\ NO_2 + O_2$

(a) 2,4 (b) 2,2 (c) 2, 3 (d) 4, 2

Q.57. Acid present in tomato is: (Set- 31/4/1)

(a) Methanoic acid (b) Acetic acid (c) Lactic acid (d) Oxalic acid

Q.58. Sodium hydroxide is termed an alkali while Ferric hydroxide is not because: (Set- 31/4/1)

(a) Sodium hydroxide is a strong base, while Ferric hydroxide is a weak base.

(b) Sodium hydroxide is a base which is soluble in water while Ferric hydroxide is also a base but it is not soluble in water.

(c) Sodium hydroxide is a strong base while Ferric hydroxide is a strong acid.

(d) Sodium hydroxide and Ferric hydroxide both are strong base but the solubility of Sodium hydroxide in water is comparatively higher than that of Ferric hydroxide.

Q.59. The name of the salt used to remove permanent hardness of water is : (Set- 31/4/1)

(a) Sodium hydrogen carbonate ($NaHCO_3$)

(b) Sodium chloride (NaCl)

(c) Sodium carbonate decahydrate ($Na_2CO_3.10H_2O$)

(d) Calcium sulphate hemihydrate ($CaSO_4.\frac{1}{2}H_2O$)

Q.60. Select a pair of olfactory indicators from the following: (Set- 31/5/1)

(a) Clove oil and vanilla essence (b) Onion and turmeric

(c) Clove oil and litmus solution (d) Vanilla and methyl orange

Q.61. Fresh milk has a pH of 6. To delay its curdling, a chemical substance is added to it, which is:

(a) Sodium carbonate (b) Baking powder (Set- 31/5/1)

(c) Sodium hydroxide (Caustic soda)

(d) Baking soda (Sodium hydrogen carbonate)

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Q.62. Hydronium ions are formed by the reaction between: (Set- 31/5/1)

(a) Sodium hydroxide and water (b) Calcium chloride and water

(c) Hydrogen chloride gas and water (d) Ethanol and water

Q.63. Few drops of aqueous solution of ammonium chloride are put on a universal indicator paper. The paper turns pink. (Set- 31/6/1)

Study the following table and choose the correct option.

Nature	Ammonium chloride is a salt of....	Range of pH
(a) acidic	weak acid and strong base	less than 7
(b) basic	weak acid and strong base	more than 7
(c) acidic	strong acid and weak base	less than 7
(d) basic	strong acid and strong base	7

Q.64. Two salts X and Y are dissolved in water separately. When phenolphthalein is added to these two solutions, the solution 'X' turns pink and the solution Y does not show any change in colour, therefore 'X' and Y are (Set- 31/6/1)

	(X)	(Y)
(a)	Na_2CO_3	NH_4Cl
(b)	Na_2SO_4	$NaHCO_3$
(c)	NH_4Cl	Na_2SO_4
(d)	$NaNO_3$	Na_2SO_4

Q.65. **Assertion (A):** It is advised that while diluting an acid one should add water to acid and not acid to water keeping the solution continuously stirred. (Set- 31/6/1)

Reason (R): The process of dissolving an acid into water is highly exothermic.

Section-B (two Marker Questions)

Q.66. (a) (i) A compound X which is prepared from gypsum has the property of hardening when mixed with proper quantity of water. (Set- 31/1/1)
Identify 'X' and write its chemical formula.

(ii) State the difference in chemical composition between baking soda and baking powder.

OR

(b) Write balanced chemical equation for the reaction that occurs when:

(i) blue coloured copper sulphate crystals are heated and

(ii) Sodium hydrogen carbonate is heated during cooking.

Q.67. (A) A student took a small amount of copper oxide in a conical flask and added dilute hydrochloric acid to it with constant stirring. He observed a change in colour of the solution. (Set- 31/4/1)

(i) Write the name of the compound formed and its colour.

(ii) Write a balanced chemical equation for the reaction involved.

OR

(B) The industrial process used for the manufacture of caustic soda involves electrolysis of an aqueous solution of compound 'X'. In this process, two gases 'Y' and 'Z' are liberated. 'Y' is liberated at cathode and 'Z', which is liberated at anode, on treatment with dry slaked lime forms a compound 'B'. Name X, Y, Z and B.

Q.68. (a) On heating 'X' at 373 K, it loses water molecules and becomes 'Y. 'Y' is a substance Which doctors use for supporting fractured bones in the right position.

(i) Identify X' and 'Y'. (ii) How can 'X' be reobtained from 'Y'? (Set- 31/5/1)

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OR

(b) Two solutions M and N give Red and Blue colour respectively with a universal indicator.
(i) In which solution will the hydrogen ion concentration be more? Justify your answer.
(ii) If both M and N solutions are mixed and the resultant mixture is tested with a Universal indicator, it turns green. What is the nature of the salt formed? Justify your answer.

Section-C (three Marker Questions)

Q.69. Suggest one remedial measure each to counteract the change in pH in human beings in following cases: (Set- 31/1/1)
(i) Production of too much acid in stomach during indigestion
(ii) Stung by a honey bee / nettle leaves
(b) Fresh milk has a pH of 6. When it changes into curd will its pH increase or decrease? Why?

Q.70. (i) Suggest a safe procedure of diluting a strong concentrated acid. (Set- 31/4/1)
(ii) Name the salt formed when sulphuric acid is added to sodium hydroxide and write its pH.
(iii) Dry HCl gas does not change the colour of dry blue litmus paper. Why?

Q.71. (a) Sometimes the pH of our mouth gets lower than 5.5. Why? (Set- 31/5/1)
(b) A basic salt X' is obtained by heating baking soda followed by crystallisation. Identify 'X' and state its two industrial uses.
(c) Why do copper sulphate crystals turn white on heating?

Q.72. (a) A substance 'X' is used as a building material and is insoluble in water. When it reacts with dil. HCl, it produces a gas which turns lime water milky. (Set- 31/6/1)
(i) Write the chemical name and formula of 'X'.
(ii) Write chemical equations for the chemical reactions involved in the above statements.

OR

(b) A metal 'M' on reacting with dilute acid liberates a gas 'G'. The same metal also liberates gas 'G' when reacts with a base.
(i) Write the name of gas 'G'.
(ii) How will you test the presence of this gas?
(iii) Write chemical equations for the reactions of the metal with (1) an acid and (2) a base.

Section-E (four Marker Questions)

Q.73. The teacher while conducting practicals in the laboratory divided the students into three groups and gave them various solutions to find out their pH and classify them into acidic, basic and neutral solutions. (Set- 31/2/1)

Group A Lemon juice, vinegar, colourless aerated drink
Group B - Tomato juice, coffee, ginger juice
Group C Sodium hydroxide, sodium chloride, lime water

(a) For the solutions provided, which group is/are likely to have pH value (i) less than 7, and (ii) greater than 7?
(b) List two ways of determining pH of a solution
(c) Explain, why the sour substances such as lemon juice are effective in cleaning the Tarnished copper vessels.
(c) "pH has great importance in our daily life." Justify this statement by giving two examples.

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